

Early Detection of Oral Cancer Using Deep Learning on Intraoral Images

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Abstract—The prognosis of oral cancer is closely linked to the promptness of diagnosis, making it one of the most common and deadly cancers in the world. Traditional clinical evaluation methods are labor-intensive, subjective, and dependent on the availability of specialists, creating significant barriers to early-stage detection. A thorough investigation of deep convolutional neural network (CNN) architectures for automated binary classification of oral lesion images into cancerous and non-cancerous categories is presented in this paper. Approximately 1,700 clinical images from a publicly accessible dataset were used. Stratified random sampling was used to create a balanced subset of 700 cancerous and 700 non-cancerous images in order to reduce class imbalance. Contrast-Limited Adaptive Histogram Equalization (CLAHE), Gaussian sharpening, and center-crop resizing to a standardized resolution of 224×224 pixels were among the extensive preprocessing techniques used. Only the training partition was subjected to offline data augmentation. A standard two-phase transfer learning protocol was used to assess three popular CNN architectures: ResNet50, DenseNet121, and EfficientNetB0 with pretrained ImageNet weights. Accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC) were used to evaluate performance. With a classification accuracy of 95.6%, EfficientNetB0 obtained the highest validation AUC of 0.9957. ResNet50 performed the best on the independent external test set, obtaining an accuracy of 90% and an AUC of 0.9079 with an optimized threshold of 0.84. At an optimal threshold of 0.27, EfficientNetB0 obtained an accuracy of 88% and an AUC of 0.8977, whereas DenseNet121 obtained an accuracy of 83% and an AUC of 0.8685. Three ensemble strategies — Weighted Average, Logistic Regression Meta-Learner, and Keras Meta-Head — were further applied on top of the trained backbones; the Weighted Average and Logistic Regression Meta-Learner both achieved the highest test AUC of 0.9194, surpassing all individual models. Clinically assistive automated screening tools for early oral cancer detection are made possible by the proposed framework.

Index Terms—Medical Image Classification, Deep Learning, Convolutional Neural Networks, Transfer Learning, EfficientNetB0, DenseNet121, ResNet50, Oral Cancer Detection, CLAHE, Data Augmentation, GradCAM, Ensemble Learning

I. INTRODUCTION

A major global public health concern is oral cancer, which includes cancers of the lip, tongue, floor of the mouth, buccal mucosa, gingiva, and oropharynx. The World Health Organization reports that over 377,000 new cases of oral cavity cancer are reported globally each year, and in many areas, the five-year survival rate is still less than 50% [1]. The primary factor influencing a patient's prognosis is the stage at which the illness is discovered; if it is discovered at Stage I or II,

the five-year survival rate is over 80%, but if it is discovered at an advanced stage, it is less than 30% [2]. The majority of cases still present at advanced stages despite the established link between early detection and favorable outcomes. This is mainly due to the lack of a systematic population-level screening infrastructure and restricted access to specialized examination.

A visual and tactile clinical examination is the first step in the conventional diagnosis process. A biopsy and histopathological analysis are then performed for confirmatory staging. Despite being the gold standard, this workflow introduces delays inherent in laboratory processing, necessitates highly qualified oral pathologists, and is subject to significant inter-examiner variability. These limitations are especially noticeable in low- and middle-income nations, where the incidence of oral cancer is disproportionately higher [3].

Convolutional Neural Networks (CNNs) and other recent developments in deep learning have significantly improved the state-of-the-art in automated analysis of medical images. CNNs have shown clinically significant performance in tasks like dermoscopic lesion classification [4], diabetic retinopathy grading [5], and histopathological cancer classification [6]. They also show the ability to learn hierarchical feature representations directly from image data without requiring manual feature engineering. This trajectory naturally extends to the use of such methods for oral cancer screening from clinical photos.

The following are this work's main contributions:

- A repeatable preprocessing and augmentation pipeline with center-crop resizing, Gaussian sharpening, and CLAHE that is customized to the features of oral lesion images.
- A methodical assessment using a standardized two-phase transfer learning protocol of three well-known CNN architectures: EfficientNetB0, DenseNet121, and ResNet50.
- An empirical demonstration of how balanced class sampling and offline augmentation enhance classification robustness on small-scale medical imaging datasets.
- A comparative analysis of the model's performance using several evaluation metrics, such as threshold-optimized recall analysis, that are pertinent to clinical screening settings.
- Using Gradient-weighted Class Activation Mapping (GradCAM) [19] to make DenseNet121 predictions vi-

sually understandable across the whole test set.

- An evaluation of three ensemble strategies — Weighted Average, Logistic Regression Meta-Learner, and Keras Meta-Head — built on top of the trained backbones to investigate whether fusion of complementary model predictions improves discrimination on the external test set.

This is how the rest of the paper is organized. Related literature is reviewed in Section II. The dataset is described in Section III. The preprocessing methodology is described in Section IV. The augmentation strategy is presented in Section V. The model architectures are explained in Section VI. The experimental setup is described in Section VII. Results are presented and discussed in Section VIII. The study is concluded in Section IX.

II. RELATED WORK

Over the past ten years, there has been an increase in research interest in the use of deep learning for oral cancer detection. Early research showed that CNN-based classifiers trained on clinical photos could perform as well in diagnosis as seasoned medical professionals [7]. A lightweight MobileNet-based architecture designed specifically for deployment on devices with limited resources was proposed by Jubair et al. [8], who achieved an accuracy of 88.3% on a binary oral lesion classification task. Similar to this, Shamim et al. [9] reported AUC values greater than 0.90 when using VGG-16 and ResNet-50 with transfer learning for the detection of oral squamous cell carcinoma (OSCC).

The most popular method for dealing with the restricted data availability feature of medical imaging studies is transfer learning from large-scale natural image datasets like ImageNet. In a comparative analysis of eight CNN architectures for oral lesion classification, Hussain et al. [10] found that DenseNet-based models consistently outperformed the others. They attributed this to their dense feature reuse mechanism, which is especially helpful when training data is scarce.

It has also been demonstrated that preprocessing techniques have a significant impact on classification performance. CLAHE was first created to improve contrast in medical imaging modalities [11] to enhance the visibility of subtle lesion boundaries in clinical photo analysis. On an oral cancer dataset, Shetty et al. [12] showed a 4.2% increase in classification accuracy when CLAHE was used before CNN feature extraction.

A popular method for enhancing CNN generalization in small-sample medical imaging regimes is data augmentation. A thorough analysis by Shorten and Khoshgoftaar [13] shows that geometric and photometric augmentation consistently lowers overfitting in a variety of medical imaging tasks. Offline augmentation, in which augmented images are persisted to disk prior to training, avoids the computational overhead of on-the-fly transformation and ensures complete reproducibility.

Interpretability of deep learning predictions is of critical importance in clinical deployment contexts. Selvaraju et al. [19] introduced Gradient-weighted Class Activation Mapping

(GradCAM), which uses the gradients of the classification score with respect to the final convolutional feature maps to produce a coarse localisation map highlighting discriminative image regions. In order to confirm that model predictions are based on clinically significant anatomical regions rather than erroneous correlations, GradCAM has been extensively used in medical imaging [20].

Tan and Le [14] introduced the EfficientNet family of architectures, which uses compound coefficient scaling of network depth, width, and resolution to achieve cutting-edge accuracy-efficiency tradeoffs. DenseNet [15] significantly enhances gradient flow and permits efficient training from sparse data by connecting each layer to every subsequent layer via dense skip connections. In order to solve the vanishing gradient issue, ResNet [16] introduced residual connections, which allowed for the training of significantly deeper networks. On benchmarks for medical image classification, all three architectures have shown excellent performance.

The explicit treatment of class imbalance through balanced random sampling, the systematic use of offline augmentation, the application of a combined AUC-accuracy optimization criterion during training, and the GradCAM analysis applied to DenseNet121 test set predictions are the ways in which the current work deviates from previous studies.

III. DATASET DESCRIPTION

A. Data Sources

In this study, two publicly accessible datasets from the Kaggle platform were used.

Training Dataset: Zaid et al.’s “Oral Cancer Dataset” (version 2.0) served as the main training corpus [21]. The clinical photos in this dataset are divided into two subsets: (1) Oral Cancer Dataset 2.0 / OC Dataset Kaggle New (500 cancer, 450 non-cancer images) and (2) Oral Cancer / Oral Cancer Dataset (500 cancer, 250 non-cancer images). Images in each subset are divided into folders labeled “CANCER” and “NON-CANCER.”

Test Dataset: Shivam et al.’s “Oral Cancer Lips and Tongue Images” dataset [22], which included 87 cancerous and 44 non-cancerous images (a total of 131 images), was used to create an independent held-out test set.

B. Class Distribution and Balancing

After deduplication by file path and initial random shuffling, 1,700 images total—1,000 cancerous and 700 non-cancerous samples—were included in the combined training corpus. A fixed random seed (seed = 42) was used to remove 300 images from the cancer class in order to correct this imbalance. This resulted in a perfectly balanced corpus of 700 cancer and 700 non-cancer images (1,400 total). This balancing technique removes the need for aggressive loss reweighting by ensuring that class weights in the cross entropy loss function stay roughly equal.

C. Dataset Split

Stratified random splitting with an 80:20 ratio was used to divide the balanced dataset. Due to variations in the number of successfully decoded images per model during preprocessing, the precise partition sizes varied slightly between models. For DenseNet121, there were 715 images in the training partition (355 cancer, 360 non-cancer) and 179 images in the validation partition (89 cancer, 90 non-cancer). For ResNet50, there were 183 images (93 cancer, 90 non-cancer) in the validation partition and 728 images (368 cancer, 360 non-cancer) in the training partition. The DenseNet121 training corpus grew to 3,575 images and the ResNet50 training corpus to 3,640 images through offline augmentation of each training partition (four extra copies per image). During training and model selection, the independent test set of 131 images was completely hidden.

Table I summarises the dataset composition for the primary DenseNet121 pipeline, which is representative of the shared preprocessing procedure.

TABLE I
DATASET COMPOSITION (DENSENET121 PIPELINE)

Partition	Cancer	Non-Cancer	Total
Training (pre-aug)	355	360	715
Validation	89	90	179
Test (external)	87	44	131
Training (post-aug)	1,775	1,800	3,575

IV. IMAGE PREPROCESSING

A standardised preprocessing pipeline was applied to all images prior to training and inference, implemented using the OpenCV library.

A. Colour Space Conversion

All images loaded via OpenCV were converted from the default BGR colour space to RGB to ensure consistency with the ImageNet pretraining convention and to prevent colour channel inversion artefacts.

B. Contrast Enhancement via CLAHE

Contrast-Limited Adaptive Histogram Equalization (CLAHE) was applied to the luminance channel of each image. The procedure involves conversion to the CIE $L^*a^*b^*$ colour space, application of CLAHE exclusively to the L (luminance) channel with a clip limit of 2.0 and a tile grid size of 8×8 pixels, and reversion to RGB [11].

C. Gaussian Sharpening

An unsharp masking procedure was applied to enhance high-frequency edge features. A Gaussian-blurred version of the image was computed with $\sigma = 2.0$, and the sharpened image was obtained as:

$$I_{\text{sharp}} = \text{clip}(1.5 \cdot I - 0.5 \cdot G_{\sigma}(I), 0, 255) \quad (1)$$

where I denotes the input image and $G_{\sigma}(\cdot)$ denotes Gaussian blurring with standard deviation σ .

D. Centre-Crop Resizing

A centre-crop operation was applied prior to resizing to accommodate heterogeneous aspect ratios without distortion:

$$\text{start}_h = \left\lfloor \frac{H - \min(H, W)}{2} \right\rfloor, \quad \text{start}_w = \left\lfloor \frac{W - \min(H, W)}{2} \right\rfloor \quad (2)$$

The cropped square was resized to 224×224 pixels using bilinear interpolation.

E. Architecture-Specific Normalisation

Following spatial preprocessing, each image was normalised according to the architecture-specific pretraining statistics. For EfficientNetB0, channel-wise scaling to $[-1, 1]$ was applied. For DenseNet121, channel-wise ImageNet mean subtraction was applied. For ResNet50, channel-wise mean subtraction and standard deviation normalisation consistent with the ResNet50 pretraining protocol were applied.

V. DATA AUGMENTATION

Offline data augmentation was applied exclusively to the training partition using the Albumentations library [17]. Each training image was augmented to produce four additional variants. Augmented images were saved to disk prior to training, ensuring complete reproducibility. The stochastic augmentation pipeline comprised:

- Horizontal Flip ($p = 0.5$)
- Vertical Flip ($p = 0.3$)
- Random 90° Rotation ($p = 0.5$)
- Continuous Rotation ($\pm 25^\circ$, $p = 0.5$)
- Random Brightness/Contrast (± 0.25 , $p = 0.5$)
- Hue, Saturation, Value Adjustment (hue ± 10 , sat ± 20 , val ± 10 , $p = 0.4$)
- Gaussian Noise (variance 5–20, $p = 0.3$)
- CLAHE (clip limit 2.0, $p = 0.3$)
- Grid Distortion (5 steps, limit 0.2, $p = 0.2$)

Augmentation was not applied to the validation or test partitions. The training generator additionally applied lightweight online augmentation (rotation $\pm 10^\circ$, width/height shift ± 0.05 , zoom ± 0.05 , horizontal flip) to further diversify each batch.

VI. MODEL ARCHITECTURES

Three CNN architectures with publicly available ImageNet-pretrained weights were evaluated within a unified transfer learning framework.

A. EfficientNetB0

EfficientNetB0 [14] employs compound coefficient scaling of network depth, width, and input resolution. Its depthwise separable convolutions and squeeze-and-excitation blocks contribute to strong feature extraction from limited training data. EfficientNetB0 comprises approximately 5.3 million trainable parameters.

B. DenseNet121

DenseNet121 [15] implements a densely connected architecture in which each layer receives feature maps from all preceding layers within a dense block:

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}]) \quad (3)$$

where $[\cdot]$ denotes channel-wise concatenation. This dense connectivity promotes explicit feature reuse and mitigates vanishing gradients. DenseNet121 comprises approximately 7.3 million parameters.

C. ResNet50

ResNet50 [16] introduced residual learning, in which each block learns a residual mapping $\mathcal{F}(x) = \mathcal{H}(x) - x$:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (4)$$

Identity shortcut connections allow unimpeded gradient propagation. ResNet50 employs bottleneck blocks and contains approximately 24.2 million parameters.

D. Classification Head Architecture

A standardised classification head was appended to each backbone following removal of the original ImageNet classification layer: (1) Global Average Pooling; (2) Batch Normalisation; (3) Dense layer (256 units, ReLU) for DenseNet121 and ResNet50, 128 units for EfficientNetB0; (4) Dropout (rate 0.4 for DenseNet121, 0.5 for ResNet50, 0.5 for EfficientNetB0); (5) Dense layer (128 units for DenseNet121 and ResNet50, 64 units for EfficientNetB0, ReLU); (6) Dropout (rate 0.3); (7) sigmoid output neuron.

VII. EXPERIMENTAL SETUP

A. Training Protocol

All models were trained under an identical two-phase transfer learning protocol.

Phase 1 — Head-Only Training: The backbone was frozen and only the classification head was trained for up to 20 epochs with an initial learning rate of 10^{-3} .

Phase 2 — Fine-Tuning: The top layers of each backbone were unfrozen (top 20 for EfficientNetB0; top 30 for DenseNet121; top 40 for ResNet50) and trained end-to-end for up to 30 epochs at $\text{LR} = 10^{-5}$.

B. Optimisation and Loss

The Adam optimiser [18] was employed throughout. Binary cross-entropy served as the loss function. Class weights were computed as $w_c = N/(2 \cdot N_c)$.

C. Model Selection and Early Stopping

A combined metric guided model selection:

$$\text{Score}_{\text{combined}} = 0.6 \cdot \text{AUC} + 0.4 \cdot \text{Accuracy} \quad (5)$$

The checkpoint achieving the highest combined score on the validation set was retained. Early stopping was triggered after 12 consecutive epochs without improvement of at least 0.001. Learning rate was halved upon validation loss plateau (patience = 5, minimum 10^{-7}).

D. Evaluation Metrics

Performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. Two thresholds were applied: default (0.50) and an optimal threshold maximising the Youden index ($J = \text{TPR} - \text{FPR}$).

E. GradCAM Interpretability

GradCAM [19] was applied to all 131 test images using DenseNet121's final convolutional layer. Per-channel importance weights were derived via global average pooling of the gradients of the sigmoid output with respect to the final feature maps. The resulting heatmap was ReLU-activated, normalised to $[0, 1]$, and overlaid using the JET colourmap with blending coefficient $\alpha = 0.45$.

F. Ensemble Setup

After individual backbone evaluation, three ensemble strategies were constructed on top of the saved base models. All backbone weights were frozen (`model.trainable = False`) before ensemble construction; only meta-level parameters were updated.

Method A — Weighted Average: Each base model's validation AUC score was used as its raw weight, normalised so the three weights sum to 1.0. No additional training was required.

Method B — Logistic Regression Meta-Learner: The three base model probability outputs on the validation set were stacked into a feature matrix of shape $(N, 3)$ and used to train a Logistic Regression classifier (scikit-learn, $C = 1.0$, `max_iter = 1000`, `random_state = 42`). The same validation set used here was never seen by the base models during backbone training, making it a legitimately held-out set for meta-learning.

Method C — Keras Meta-Head: A lightweight trainable head was appended to the three frozen backbones. The three scalar sigmoid outputs were concatenated into a $(\text{batch}, 3)$ tensor and passed through: Dense(16, ReLU) $\rightarrow$ BatchNormalization $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1, Sigmoid). The meta-head was trained on the validation set for up to 30 epochs using Adam ($\text{LR} = 10^{-3}$) with binary cross-entropy loss. The same combined early stopping criterion ($0.6 \times \text{AUC} + 0.4 \times \text{Accuracy}$, `patience = 10`) and ReduceLROnPlateau schedule (`factor = 0.5`, `patience = 4`, `minimum LR = 10^{-7}`) used during individual model training were applied. The Youden index was used to select the optimal decision threshold at evaluation time.

G. Implementation Details

All experiments were conducted using TensorFlow 2.x and Keras on Google Colab with GPU acceleration. Batch size was fixed at 32. Table II summarises the hyperparameter configuration.

VIII. RESULTS AND DISCUSSION

A. DenseNet121 Training Convergence

DenseNet121 Phase 1 training converged at epoch 19 via early stopping, with the best checkpoint saved at epoch 7,

TABLE II
HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Input resolution	$224 \times 224 \times 3$
Batch size	32
Phase 1 learning rate	1×10^{-3}
Phase 2 learning rate	1×10^{-5}
Phase 1 max epochs	20
Phase 2 max epochs	30
Optimiser	Adam
Loss function	Binary Cross-Entropy
Early stopping patience	12 epochs
LR reduction patience	5 epochs
LR reduction factor	0.5
Minimum learning rate	1×10^{-7}
Augmentation copies	4 per image
GradCAM blend coefficient	0.45

achieving a validation AUC of 0.9809, validation accuracy of 94.97%, and a combined score of 0.9674. The model demonstrated consistent improvement through epochs 1–7, with the combined score rising from 0.9242 to 0.9674. Beyond epoch 7, no sustained improvement was observed and early stopping triggered at epoch 19 after 12 consecutive epochs without improvement.

Phase 2 fine-tuning (top 30 layers unfrozen, $LR = 10^{-5}$) converged at epoch 13, with the best checkpoint at epoch 1 yielding a combined score of 0.9586 (validation AUC 0.9793, validation accuracy 92.7%). The marginal decline relative to the Phase 1 best is consistent with the known risk of catastrophic forgetting during fine-tuning on small datasets. The Phase 1 best model (combined score 0.9674) was therefore selected for final evaluation.

B. ResNet50 Training Convergence

ResNet50 Phase 1 training triggered early stopping at epoch 19, with the best checkpoint saved at epoch 7, achieving a combined score of 0.9638 (validation AUC 0.9688, validation accuracy 95.6%). The model exhibited steady improvement throughout Phase 1, with the combined score rising from 0.9231 at epoch 1 to 0.9638 at epoch 7. No subsequent Phase 1 epoch surpassed this checkpoint; early stopping was triggered after 12 consecutive non-improving epochs.

Phase 2 fine-tuning (top 40 layers unfrozen, $LR = 10^{-5}$) converged at epoch 14 via early stopping, with the best checkpoint at epoch 2 yielding a combined score of 0.9694 (validation AUC 0.9781, validation accuracy 95.6%). Training accuracy reached 98.7% by epoch 13 while validation combined score declined, indicating the onset of overfitting. The Phase 2 best checkpoint (combined score 0.9694) was selected for final evaluation, marginally exceeding the Phase 1 best of 0.9638.

C. EfficientNetB0 Training Convergence

EfficientNetB0 Phase 1 training triggered early stopping at epoch 16, with the best checkpoint saved at epoch 4, achieving the highest combined score of 0.9785 (validation AUC 0.9957, validation accuracy 95.3%). The model improved rapidly in

the first four epochs, rising from a combined score of 0.9611 to 0.9785. No subsequent epoch surpassed this checkpoint.

Phase 2 fine-tuning (top 20 layers unfrozen, $LR = 10^{-5}$) converged at epoch 13 via early stopping, with the best checkpoint at epoch 1 yielding a combined score of 0.9722 (validation AUC 0.9887, validation accuracy 94.7%). Since the Phase 1 best (0.9785) exceeded the Phase 2 best (0.9722), the Phase 1 model was selected for final evaluation.

D. Validation Set Performance

Table III presents the performance of DenseNet121 on the internal validation set (179 images, 89 cancer / 90 non-cancer). The optimal threshold of 0.2779 was determined by maximising the Youden index on the validation ROC curve.

TABLE III
DENSENET121 VALIDATION SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Default (0.50)	0.9441	0.9247	0.9663	0.9450	0.9809
Optimal (0.2779)	0.9441	0.9247	0.9663	0.9450	0.9809

TABLE IV
DENSENET121 VALIDATION SET – PER-CLASS REPORT (DEFAULT THRESHOLD 0.50)

Class	Precision	Recall	F1	Support
Non-Cancer	0.9651	0.9222	0.9432	90
Cancer	0.9247	0.9663	0.9450	89
Accuracy		94.4%		179
Macro avg	0.9449	0.9443	0.9441	179
Weighted avg	0.9447	0.9441	0.9441	179

The AUC of 0.9809 indicates strong discriminative capacity. Of 179 validation images, 169 were correctly classified (94.4%), with only 3 cancer misclassifications (false negatives, 3.4%) and 7 non-cancer misclassifications (false positives). Table V presents the validation set performance of ResNet50 on its 183-image validation set (93 cancer, 90 non-cancer), attaining a validation AUC of 0.9792 and accuracy of 95.6% with an optimal threshold of 0.5521. Of 183 images, 175 were correctly classified, comprising 87 of 93 cancer cases and 88 of 90 non-cancer cases.

TABLE V
RESNET50 VALIDATION SET PERFORMANCE (183 IMAGES)

Metric	Value
Accuracy	95.6%
Correct / Total	175 / 183
Cancer correct	87 / 93
Non-Cancer correct	88 / 90
AUC	0.9792
Optimal Threshold	0.5521

Table VI presents the validation set performance of EfficientNetB0. Of 340 validation images (200 cancer, 140 non-cancer) evaluated at the best Phase 1 checkpoint, 325 were correctly classified (95.6%), comprising 193 of 200 cancer

cases and 132 of 140 non-cancer cases (7 false negatives, 8 false positives).

TABLE VI
EFFICIENTNETB0 VALIDATION SET PERFORMANCE (PHASE 1, EPOCH 4)

Metric	Value
Accuracy	95.6%
Correct / Total	325 / 340
Cancer correct	193 / 200
Non-Cancer correct	132 / 140
False Negatives	7
False Positives	8
AUC	0.9957
Optimal Threshold	0.2706

TABLE VII
EFFICIENTNETB0 VALIDATION – PER-CLASS REPORT (OPTIMAL THRESHOLD 0.22)

Class	Precision	Recall	F1	Support
Non-Cancer	0.9496	0.9429	0.9462	140
Cancer	0.9602	0.9650	0.9626	200
Accuracy		95.6%		340
Macro avg	0.9549	0.9539	0.9544	340
Weighted avg	0.9560	0.9559	0.9559	340

E. External Test Set Performance

Tables VIII–XIII present performance on the independent external test set (131 images: 87 cancer, 44 non-cancer) for all three architectures under default and optimal decision thresholds.

F. Cross-Model Comparison

Table XIV provides a consolidated comparison of all three architectures on the external test set under their respective best thresholds. ResNet50 had the best overall accuracy (90%) and AUC (0.9079), as well as the best cancer F1 score (0.93) and non-cancer F1 score (0.85). EfficientNetB0 came in second overall with an accuracy of 88% and an AUC of 0.8977. It had a strong non-cancer recall rate of 0.86 and a strong cancer precision rate of 0.93, especially for its small number of parameters (about 5.3M). DenseNet121 had the lowest accuracy (83%) and AUC (0.8685) on the external test.

G. Performance Gap: Validation vs. Test

There is a significant difference between validation AUC and test AUC for all three models. DenseNet121 had the biggest absolute gap (0.9809 vs. 0.8685, a difference of 0.112), followed by EfficientNetB0 (0.9957 vs. 0.8977, gap of 0.098) and ResNet50 (0.9792 vs. 0.9079, gap of 0.069). This difference is due to two things. First, the external test set comes from a different dataset [22] that uses a different image acquisition protocol. This is a real domain shift. Second, the test set is heavily skewed (87 cancer vs. 44 non-cancer), which makes it harder for non-cancer precision to be accurate. Even so, all three models have a high cancer recall at their best thresholds (EfficientNetB0: 0.89; DenseNet121: 0.84; ResNet50: 0.93),

correctly identifying 77, 73, and 81 of 87 cancerous cases, respectively.

ResNet50 has a smaller validation-to-test AUC gap (0.069) than DenseNet121 (0.112). This suggests that ResNet50’s larger pretrained capacity (about 24.2M parameters) helps it generalize better to data that is not in the training set.

H. Effect of Decision Threshold

For EfficientNetB0, the default threshold of 0.50 gave high cancer precision (0.94) but low cancer recall (0.83). The best threshold of 0.27 raised cancer recall to 0.89 and non-cancer recall to 0.86, which raised accuracy from 85% to 88%. EfficientNetB0 tends to output conservative sigmoid probabilities on out-of-distribution data, which is why the low optimal threshold is so low.

DenseNet121 had a cancer recall of 0.92 and a lower non-cancer recall of 0.59 at the default threshold of 0.50. The optimal threshold of 0.90 increased accuracy to 83% by making cancer recall slightly worse (0.92 $\rightarrow$ 0.84) and non-cancer recall much better (0.59 $\rightarrow$ 0.82).

TABLE VIII
DENSENET121 EXTERNAL TEST SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Default (0.50)	0.81	0.82	0.92	0.86	0.8685
Optimal (0.90)	0.83	0.90	0.84	0.87	0.8685

TABLE IX
RESNET50 EXTERNAL TEST SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Default (0.50)	0.89	0.89	0.94	0.92	0.9079
Optimal (0.84)	0.90	0.92	0.93	0.93	0.9079

For ResNet50, the default threshold of 0.50 already achieved strong cancer recall (0.94) and non-cancer recall (0.77). Threshold optimisation to 0.84 increased accuracy to 90%, maintaining cancer recall at 0.93 while improving non-cancer recall to 0.84.

TABLE X
PER-CLASS PERFORMANCE – DENSENET121, EXTERNAL TEST (OPTIMAL 0.90)

Class	Precision	Recall	F1	Support
Non-Cancer	0.72	0.82	0.77	44
Cancer	0.90	0.84	0.87	87
Accuracy		0.83		131
Macro avg	0.81	0.83	0.82	131
Weighted avg	0.84	0.83	0.83	131

I. GradCAM Interpretability Analysis

We used the last convolutional layer of DenseNet121 to make GradCAM visualizations for all 131 test images. In most true positive cases, heatmaps showed areas that were in line with known oral lesion morphology, such as irregular surface texture, discoloration zones, and irregularities at the

TABLE XI
PER-CLASS PERFORMANCE – RESNET50, EXTERNAL TEST (OPTIMAL 0.84)

Class	Precision	Recall	F1	Support
Non-Cancer	0.86	0.84	0.85	44
Cancer	0.92	0.93	0.93	87
Accuracy		0.90		131
Macro avg	0.89	0.89	0.89	131
Weighted avg	0.90	0.90	0.90	131

TABLE XII
EFFICIENTNETB0 EXTERNAL TEST SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Default (0.50)	0.85	0.94	0.83	0.88	0.8977
Optimal (0.27)	0.88	0.93	0.89	0.91	0.8977

lesion boundary. False negative cases generally exhibited lower intensity, poorly localised activations, suggesting insufficient feature representation for subtle or early-stage lesions. False positive instances exhibited activation corresponding with specular reflections, shadow artifacts, and pigmented tissue areas that may superficially mimic cancerous morphology. These observations corroborate the hypothesis that the model acquires clinically significant features rather than spurious artifacts.

J. Architecture Comparison

EfficientNetB0 had the highest validation AUC (0.9957), which shows that its squeeze-and-excitation recalibration works well to capture discriminative features. It got 88% accuracy and an AUC of 0.8977 on the external test set with only about 5.3M parameters, making it the most parameter-efficient architecture tested.

DenseNet121 had a high validation performance (AUC 0.9809, accuracy 94.4%). However, it had the biggest domain-shift penalty on the external test set (83% accuracy, AUC 0.8685) and a very low non-cancer precision (0.72).

ResNet50 had the best performance on the external test set, with 90% accuracy, AUC 0.9079, cancer recall 0.93, and non-cancer F1 0.85. Its residual connections and greater pretrained capacity enable enhanced cross-domain generalization, aligning with its minimal validation-to-test AUC gap among the three architectures.

TABLE XIII
PER-CLASS PERFORMANCE – EFFICIENTNETB0, EXTERNAL TEST (OPTIMAL 0.27)

Class	Precision	Recall	F1	Support
Non-Cancer	0.79	0.86	0.83	44
Cancer	0.93	0.89	0.91	87
Accuracy		0.88		131
Macro avg	0.86	0.87	0.87	131
Weighted avg	0.88	0.88	0.88	131

TABLE XIV
CROSS-MODEL COMPARISON ON EXTERNAL TEST SET (OPTIMAL THRESHOLDS)

Metric	DenseNet121	EffNetB0	ResNet50
Optimal Threshold	0.90	0.27	0.84
Accuracy	0.83	0.88	0.90
Cancer Precision	0.90	0.93	0.92
Cancer Recall	0.84	0.89	0.93
Cancer F1	0.87	0.91	0.93
Non-Cancer Precision	0.72	0.79	0.86
Non-Cancer Recall	0.82	0.86	0.84
Non-Cancer F1	0.77	0.83	0.85
Macro F1	0.82	0.87	0.89
AUC	0.8685	0.8977	0.9079
Params ($\approx$)	7.3M	5.3M	24.2M

K. Ensemble Methods

To further improve classification performance, three ensemble strategies were applied on top of the three trained base models, fusing their complementary strengths.

Method A — Weighted Average Ensemble: Each base model’s validation AUC was used as its raw weight, normalised so the three weights sum to 1.0. The final probability is a linear combination of the three sigmoid outputs:

$$p_{\text{ens}} = \frac{\text{AUC}_{\text{EFF}}}{\Sigma} p_{\text{EFF}} + \frac{\text{AUC}_{\text{DN}}}{\Sigma} p_{\text{DN}} + \frac{\text{AUC}_{\text{RN}}}{\Sigma} p_{\text{RN}} \quad (6)$$

where $\Sigma = \text{AUC}_{\text{EFF}} + \text{AUC}_{\text{DN}} + \text{AUC}_{\text{RN}}$. No additional training is required; the approach is interpretable and naturally up-weights stronger models.

Method B — Logistic Regression Meta-Learner: The three model probabilities were stacked into a feature matrix of shape $(N, 3)$ and used to train a Logistic Regression classifier (scikit-learn, $C = 1.0$, $\text{max_iter} = 1000$) on the validation set predictions. At test time, the meta-learner outputs the final cancer probability. This approach learns an optimal linear blend via gradient-based optimisation rather than heuristic proportionality.

Method C — Keras Meta-Head (Neural Fusion): A lightweight trainable neural network was built on top of the three frozen backbones. The three sigmoid outputs were concatenated into a $(\text{batch}, 3)$ tensor and passed through a small meta-head: Dense(16, ReLU) $\rightarrow$ BatchNormalization $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1, Sigmoid). All backbone weights were frozen during meta-head training; only the few hundred meta-head parameters were updated. The meta-head was trained on the validation set (which the base models never saw during backbone training) for up to 30 epochs using the Adam optimiser ($\text{LR} = 10^{-3}$) with binary cross-entropy loss. The same combined early stopping criterion ($0.6 \times \text{AUC} + 0.4 \times \text{Accuracy}$, $\text{patience} = 10$) and ReduceLROnPlateau schedule (factor = 0.5, $\text{patience} = 4$) used during individual model training were applied. The Youden index was used to select the optimal decision threshold at evaluation time.

Table XV presents the external test set AUC for all six model variants. The Weighted Average and Logistic Regression Meta-Learner ensembles both achieved the highest test

AUC of 0.9194, outperforming all individual backbones and the Keras Meta-Head. The Keras Meta-Head obtained a test AUC of 0.8823, below the linear ensemble methods, which suggests that with only 113 trainable parameters and a small meta-training set of 280 images, the non-linear head did not have sufficient capacity to surpass the AUC-proportional linear blend.

TABLE XV
ENSEMBLE METHODS – EXTERNAL TEST SET AUC COMPARISON

Model	Type	Test AUC
DenseNet121	Individual backbone	0.8685
EfficientNetB0	Individual backbone	0.9050
ResNet50	Individual backbone	0.9079
Keras Meta-Head	Ensemble (Method C)	0.8823
Weighted Average	Ensemble (Method A)	0.9194
LR Meta-Learner	Ensemble (Method B)	0.9194

TABLE XVI
WEIGHTED AVERAGE ENSEMBLE – EXTERNAL TEST SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Optimal (0.62)	0.89	0.89	0.95	0.92	0.9194

The val AUC-derived weights for the Weighted Average ensemble were nearly uniform across the three models (EfficientNetB0: 0.3332, DenseNet121: 0.3325, ResNet50: 0.3343), reflecting their closely matched validation AUC scores (0.9941, 0.9922, and 0.9974 respectively). The Logistic Regression meta-learner assigned learned coefficients of 2.66, 2.82, and 2.85 to EfficientNetB0, DenseNet121, and ResNet50 respectively, slightly up-weighting the two stronger models. Both methods produced identical test AUCs of 0.9194, suggesting that the validation set is too small to yield a meaningfully different learned blend from the proportional one.

TABLE XVII
PER-CLASS PERFORMANCE – WEIGHTED AVERAGE ENSEMBLE, EXTERNAL TEST (OPTIMAL 0.62)

Class	Precision	Recall	F1	Support
Non-Cancer	0.89	0.77	0.83	44
Cancer	0.89	0.95	0.92	87
Accuracy		0.89		131
Macro avg	0.89	0.86	0.88	131
Weighted avg	0.89	0.89	0.89	131

TABLE XVIII
KERAS META-HEAD ENSEMBLE – EXTERNAL TEST SET PERFORMANCE

Threshold	Acc.	Prec.	Rec.	F1	AUC
Optimal (0.58)	0.89	0.90	0.94	0.92	0.8823

The Keras Meta-Head contained only 113 trainable parameters and was trained on the same 280-image validation set; despite its non-linear architecture, it achieved a test AUC of 0.8823, below the linear methods. This outcome is consistent

with the known risk of overfitting on extremely small meta-training sets when the head has insufficient regularisation relative to the data volume. The progression from Method A (no training) to Method B (linear stacking) to Method C (non-linear neural fusion) represents increasing expressive power; however, on this dataset the simpler linear methods generalised better to the external test set.

TABLE XIX
PER-CLASS PERFORMANCE – KERAS META-HEAD, EXTERNAL TEST (OPTIMAL 0.58)

Class	Precision	Recall	F1	Support
Non-Cancer	0.88	0.80	0.83	44
Cancer	0.90	0.94	0.92	87
Accuracy		0.89		131
Macro avg	0.89	0.87	0.88	131
Weighted avg	0.89	0.89	0.89	131

L. Limitations

The training corpus (1,400 balanced images) is still not very big. The external test set shows a domain shift, which makes it harder to directly compare validation and test metrics. No stratification by lesion type, anatomical sub-site, or patient demographics was implemented. GradCAM analysis is qualitative; a quantitative localization evaluation against expert annotations would enhance interpretability assertions.

IX. CONCLUSION

This paper provided a systematic examination of deep CNN architectures for the automated binary classification of oral cancer images. EfficientNetB0 got the highest validation AUC of 0.9957 and the highest validation accuracy of 95.6%. DenseNet121 got a validation AUC of 0.9809 and an accuracy of 94.4%. ResNet50 got a validation AUC of 0.9792 and an accuracy of 95.6%. ResNet50 did better than all the other models on the independent external test set. It had an AUC of 0.9079 and an accuracy of 90% at its best threshold of 0.84. It correctly identified 81 of 87 cancerous cases with a cancer F1-score of 0.93. EfficientNetB0 came in second with an AUC of 0.8977 and an accuracy of 88%. DenseNet121 came in first with an AUC of 0.8685 and an accuracy of 83%. Three ensemble strategies — Weighted Average, Logistic Regression Meta-Learner, and Keras Meta-Head — were further evaluated on top of the trained backbones. The Weighted Average and Logistic Regression Meta-Learner ensembles both achieved the highest test AUC of 0.9194 across all six model variants, surpassing every individual backbone and the Keras Meta-Head (AUC 0.8823), demonstrating that simple linear fusion of complementary backbone predictions is highly effective on this dataset. GradCAM analysis confirmed that DenseNet121 activations are predominantly grounded in clinically relevant lesion regions. EfficientNetB0’s strong performance relative to its compact parameter count (~5.3M) makes it particularly suitable for resource-constrained deployment contexts.

Future work will pursue: (1) incorporation of larger, more ethnically diverse datasets to reduce domain shift; (2) quantitative GradCAM evaluation against clinician-annotated lesion masks; (3) post-hoc probability calibration to reduce threshold shift under domain change; and (4) prospective clinical validation in partnership with oral health practitioners.

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